Inversion and projective methods

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L5-GE1 Level 5 Advanced · Geometry

Inversion, cross-ratio, harmonic division, poles and polars, Brocard's theorem and projective maps, with complete proofs.
By the end you can choose a centre and radius of inversion and translate a problem through it; recognise and prove harmonic ranges, and use poles, polars and Brocard's theorem; use a projective map to send a line to infinity and write the argument rigorously.
Prerequisites: Olympiad angle chasing (L4-GE1), Lengths and ratios (L4-GE2). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Prove that for any four points A, B, C, D in the plane, $AB \cdot CD + AD \cdot BC \geq AC \cdot BD$. (Ptolemy's inequality.) There is no circle in the statement and the terms are products of lengths. Invert about A with radius 1 (defined below). Theorem 1.1 gives $B'C' = BC/(AB \cdot AC)$, and similarly for the pairs C, D and B, D, so the triangle inequality $B'C' + C'D' \geq B'D'$ reads

$$\frac{BC}{AB \cdot AC} + \frac{CD}{AC \cdot AD} \geq \frac{BD}{AB \cdot AD}.$$

Multiplying by $AB \cdot AC \cdot AD$ gives the result. Equality holds if and only if C' lies on segment $B'D'$, and by Theorem 1.2 this happens if and only if A, B, C, D lie on a circle (or a line) with C on the arc BD not containing A.

This unit covers inversion, which turns circles through a chosen point into lines, then cross-ratio, harmonic division and polars, and finally projective maps, which send a chosen line to infinity. Power of a point, radical axes and directed angles (L4-GE1, L4-GE2) are assumed.

1.1 1. Inversion

Let ω be a circle with centre O and radius r. The *inverse* of a point $P \neq O$ is the point P' on ray OP with $OP \cdot OP' = r^2$. Inversion is its own inverse, fixes ω pointwise and swaps its inside and outside. We add a point at infinity and say that inversion swaps O and ∞.

Theorem 1.1.
If $P, Q \neq O$ and O, P, Q are not collinear, then triangles OPQ and $OQ'P'$ are similar, and

$$P'Q' = \frac{r^2 \cdot PQ}{OP \cdot OQ}.$$

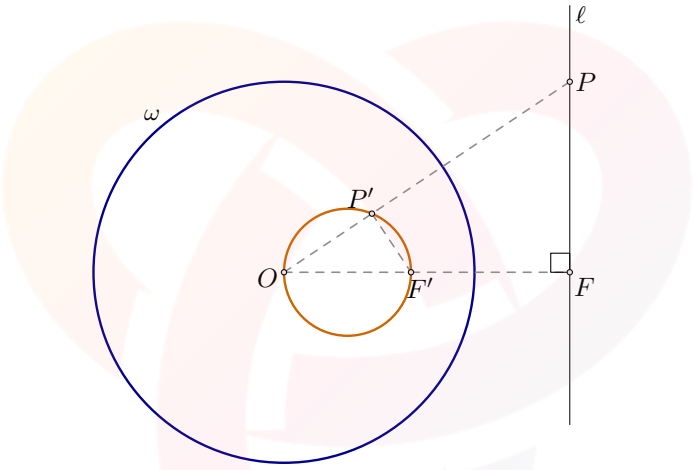
Proof. From $OP \cdot OP' = OQ \cdot OQ'$ we get $OP/OQ = OQ'/OP'$, and the angle at O is common, so $OPQ \sim OQ'P'$ with ratio $OQ'/OP = r^2/(OP \cdot OQ)$. The collinear case is a direct computation. ■

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Theorem 1.2.
(1) A line through O is mapped to itself. (2) A line not through O is mapped to a circle through O, and conversely. (3) A circle not through O is mapped to a circle not through O.

Proof. Statement (1) is in the definition. For (2), let F be the foot of the perpendicular from O to the line ℓ. For P on ℓ, Theorem 1.1 gives $\angle OP'F' = \angle OFP = 90^\circ$, so P' lies on the circle with diameter OF'; running the argument backwards shows every point of that circle other than O is the image of a point of ℓ. Its tangent at O is parallel to ℓ. For (3), let γ be the circle and $k \neq 0$ the power of O with respect to γ. For P on γ, line OP meets γ again at Q with $\overrightarrow{OP} \cdot \overrightarrow{OQ} = k$ (with $Q = P$ for a tangent). Then $\overrightarrow{OP'} = \frac{r^2}{OP^2}\overrightarrow{OP} = \frac{r^2}{k}\overrightarrow{OQ}$, so P' is the image of Q under the homothety with centre O and ratio r^2/k. Hence γ' is the image of γ under that homothety. ■

The proof of (3) shows that the centre of γ is *not* mapped to the centre of γ'; the two centres are only collinear with O.



Theorem 1.3.
Inversion preserves the angle between two curves (lines or circles) at a common point $P \neq O$. In particular tangent curves are mapped to tangent curves, and orthogonal circles to orthogonal circles.

Proof. First, two different circles or lines touch at P if and only if P is their only common point, and inversion is a bijection, so curves touching at $P \neq O$ go to curves touching at P'. Now let γ pass through P with tangent t at P, and let δ be the circle through O tangent to t at P (the line OP if t passes through O). Then δ touches γ at P (or equals it), so the line δ' is the tangent to γ' at P', and by the note in the proof of 1.2(2), δ' is parallel to the tangent to δ at O. The tangents to δ at O and at P are mirror images in the perpendicular bisector of OP, so the tangent to γ' at P' makes with line OP the angle opposite to the one t makes. Doing this for two curves through P, the angle between them keeps its size and reverses orientation. ■

1.2 2. Choosing the centre and the radius

Rules for choosing, in order of how often they apply:

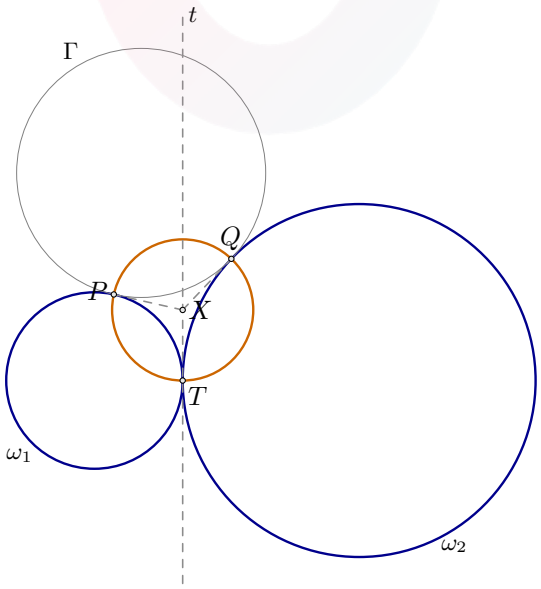
1. Invert at a point through which several circles pass; they become lines. A point where two circles touch is the best choice, as the two circles become parallel lines.

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2. If a product such as $AX \cdot AY$ is fixed, invert at A with radius $\sqrt{AX \cdot AY}$, so that X and Y swap.
3. ($\sqrt{bc}$ inversion.) In triangle ABC, invert at A with radius $\sqrt{AB \cdot AC}$ and then reflect in the bisector of angle A. This map swaps B and C, and swaps line BC with the circumcircle Γ. For example, the foot D of the altitude goes to the point of Γ on the reflection of line AD, which is line AO; so it goes to the antipode A^* of A, and $AD \cdot AA^* = AB \cdot AC$, the formula $2R \cdot AD = bc$.
4. Before solving, write down what every object becomes.

Example 1.
Circles ω_1 and ω_2 touch externally at T. A circle Γ, not through T, touches ω_1 at P and ω_2 at Q, and T, P, Q are not collinear. Prove that the circle through T, P, Q meets ω_1 and ω_2 at right angles, and that its centre lies on the common tangent of ω_1 and ω_2 at T.

Solution.
Three of the four circles pass through T, so we invert at T with any radius. The images ℓ_1, ℓ_2 of ω_1, ω_2 are lines, and they have no common point (the only common point of ω_1, ω_2 is T, which goes to ∞), so they are parallel. Each is parallel to the common tangent t at T, by the note in Theorem 1.2. The circle Γ becomes a circle Γ' touching ℓ_1 at P' and ℓ_2 at Q' (Theorem 1.3), so $P'Q'$ is a diameter of Γ' perpendicular to both lines.
The circle TPQ passes through T, so its image is the line $P'Q'$. This line does not pass through T, since otherwise T, P', Q' and hence T, P, Q would be collinear. The line $P'Q'$ meets ℓ_1 at right angles, so by Theorem 1.3 the circle TPQ meets ω_1 at right angles at P; similarly for ω_2 at Q.
For the centre X of circle TPQ: orthogonality at P means XP is tangent to ω_1, so the power of X with respect to ω_1 is XP^2. Similarly its power with respect to ω_2 is $XQ^2 = XP^2$. Hence X lies on the radical axis of ω_1 and ω_2, which is t.



1.3 3. Cross-ratio and harmonic division

For collinear points A, B, C, D define

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$$(A,B;C,D) = \frac{CA}{CB} \div \frac{DA}{DB}$$

with signed lengths. If four lines through a point P meet a line at A, B, C, D, the area formula gives $\frac{CA}{CB} = \frac{PA \sin \angle CPA}{PB \sin \angle CPB}$, and the same for D, so PA and PB cancel and $(A,B;C,D)$ depends only on the four lines (using directed angles for signs). We write $P(A,B;C,D)$ for this common value: **cross-ratio is preserved by projecting from a point onto another line.** For four points on a circle, $P(A,B;C,D)$ is the same for every P on the circle, by inscribed angles, and we call it $(A,B;C,D)$ on the circle.

We call $A,B;C,D$ *harmonic* if $(A,B;C,D) = -1$, that is, C and D divide AB internally and externally in the same ratio. Note that $(A,B;C,\infty) = CA/CB$, so $(A,B;C,\infty) = -1$ if and only if C is the midpoint of AB.

Theorem 3.1.
(1) If C, D on line AB are the feet of the internal and external bisectors from P in triangle PAB, then $(A,B;C,D) = -1$. Conversely, if $(A,B;C,D) = -1$ and $\angle CPD = 90^\circ$, then PC and PD bisect the angles between PA and PB.
(2) Let cevians AD, BE, CF of triangle ABC be concurrent, and let line EF meet line BC at T. Then $(B,C;D,T) = -1$.
(3) If $(A,B;C,D) = -1$ and M is the midpoint of AB, then $MC \cdot MD = MA^2$.
(4) If points X, Y, Z on a line have coordinates x, y, z measured from a point P of the line, then $(X,Y;P,Z) = -1$ if and only if $\frac{2}{z} = \frac{1}{x} + \frac{1}{y}$.

Statement (1) is the angle bisector theorem: $\frac{CA}{CB} = \frac{PA}{PB} = \frac{DA}{DB}$ in absolute value, with opposite signs. For the converse, the circle with diameter CD passes through P and is the Apollonius circle $\{X : XA/XB = CA/CB\}$, so $PA/PB = CA/CB$ and PC is the internal bisector. For (2), Ceva gives $\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1$ and Menelaus for the line EFT gives $\frac{BT}{TC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1$ (signed); dividing, $\frac{BD}{DC} = -\frac{BT}{TC}$. Statements (3) and (4) are short computations: for (3) put $M = 0$, $A = -1$, $B = 1$, and $(A,B;C,D) = -1$ becomes $cd = 1$; for (4), $(X,Y;P,Z) = \frac{x}{y} \cdot \frac{z-y}{z-x} = -1$ rearranges to $z(x+y) = 2xy$.

Statement (2) is usually applied to a quadrilateral. If $ABCD$ is a quadrilateral with $Q = AB \cap CD$, $R = AD \cap BC$, $P = AC \cap BD$, then in triangle QAD the cevians AC, DB and QP concur, so if QP meets AD at X, then $(A,D;X,R) = -1$. You should be able to see this harmonic range instantly.

1.4 4. Poles and polars

The *polar* of a point $P \neq O$ with respect to ω is the line through the inverse P' perpendicular to OP; P is the *pole* of that line. If P is outside ω and PT is a tangent, then $OT^2 = OP \cdot OP'$ shows that P' is the foot of T on OP, so the polar of P is the chord joining the two points of contact. The polar of a point of ω is the tangent there. With O as origin, the polar of P is the line $\{X : X \cdot P = r^2\}$ (dot product).

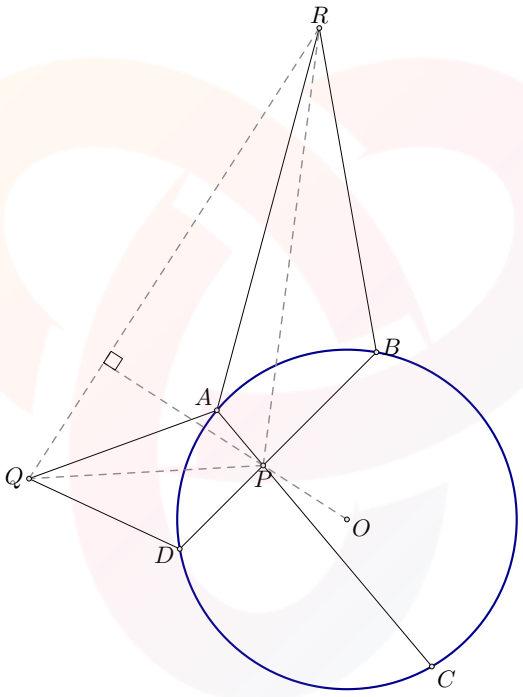
Theorem 4.1 (La Hire).
Q lies on the polar of P if and only if P lies on the polar of Q.
Theorem 4.2. If a line through P (not on ω) meets ω at X, Y and the polar of P at Z, then $(X,Y;P,Z) = -1$.
Theorem 4.3 (Brocard). Let $ABCD$ be inscribed in ω, and suppose $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$ exist. Then QR, RP, PQ are the polars of P, Q, R. Hence O is

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the orthocentre of triangle PQR.

Theorem 4.1 is trivial in coordinates: both statements say $P \cdot Q = r^2$. For 4.2, take O as origin, $r = 1$, and write the line as $P + tu$ with $|u| = 1$. The parameters t_1, t_2 of X, Y are the roots of $t^2 + 2(P \cdot u)t + |P|^2 - 1 = 0$, and Z has parameter $t_3 = (1 - |P|^2)/(P \cdot u)$. Then $\frac{1}{t_1} + \frac{1}{t_2} = \frac{-2P \cdot u}{|P|^2 - 1} = \frac{2}{t_3}$, which is Theorem 3.1(4). If $P \cdot u = 0$, then $Z = \infty$ and $t_2 = -t_1$, so again $(X, Y; P, Z) = -1$.

Proof. of Theorem 4.3. Let QP meet AD at X and BC at Y. By the quadrilateral form of Theorem 3.1(2), $(A, D; X, R) = -1$, and in the same way (triangle QBC) $(B, C; Y, R) = -1$. By Theorem 4.2 and the uniqueness of the harmonic conjugate, X and Y lie on the polar of R, so that polar is line $XY = PQ$. Similarly the polar of Q is PR. Now P lies on the polars of Q and R, so by La Hire, Q and R lie on the polar of P, which is therefore QR. Finally a polar is perpendicular to the line joining its pole to O, so $OP \perp QR$, $OQ \perp RP$, $OR \perp PQ$. ■



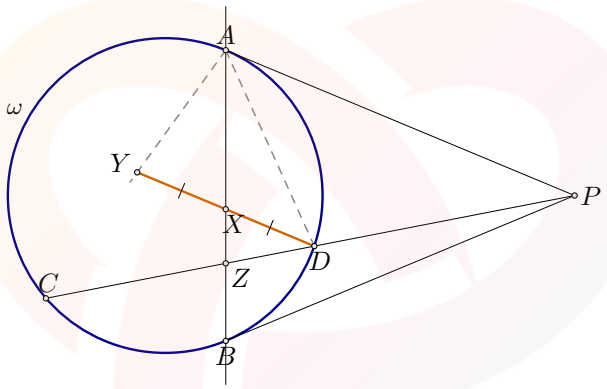
Example 2.
The incircle of triangle ABC, with centre I and $AB \neq AC$, touches BC, CA, AB at D, E, F. Line EF meets line BC at T. Prove that $(B, C; D, T) = -1$ and $IT \perp AD$.

Solution.
The point T exists: $EF \perp AI$, and $BC \perp AI$ only if $AB = AC$. Since $BD = BF$, $CD = CE$ and $AE = AF$, Ceva shows that AD, BE, CF are concurrent, and Theorem 3.1(2) gives $(B, C; D, T) = -1$.
For the perpendicularity, we should instantly spot A as the pole of EF (it is the chord of contact from A). Since T lies on the polar of A, La Hire says A lies on the polar of T. Since T lies on BC, which is the polar of D (the tangent at D), D lies on the polar of T. Hence the polar of T is AD, and it is perpendicular to IT.

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Example 3.
From a point P outside a circle ω, tangents touch ω at A and B, and a line through P meets ω at C and D. The line through D parallel to PA meets line AB at X and line AC at Y. Prove that X is the midpoint of DY.

Solution.
A midpoint and a parallel line suggest a harmonic range containing a point at infinity. Let line PCD meet AB at Z; it does meet it, because the line through P parallel to AB is perpendicular to OP at P and misses ω. Since AB is the polar of P, Theorem 4.2 gives $(C, D; P, Z) = -1$.
Now project from A onto the line m through D parallel to AP. The lines AC, AD, AP, AZ meet m at Y, D, the point at infinity, and X respectively. Hence $(Y, D; \infty, X) = -1$, so $XY = -XD$ as signed lengths and X is the midpoint of DY. Note that $C, D \neq A$, since the secant is not the tangent PA; so A is not on m, and the lines AC and AB are not parallel to m, which means X and Y exist.



Example 4.
Let $ABCD$ be a convex cyclic quadrilateral with no two sides parallel. Lines AB and CD meet at Q, and lines AD and BC meet at R. Prove that $QR^2 = QA \cdot QB + RA \cdot RD$.

Solution.
The right side is the sum of the powers of Q and R with respect to the circle ω (both points lie outside it, on extensions of chords). With the centre O as origin and r the radius, the power of Q is $|Q|^2 - r^2$, so the claim reads $|Q - R|^2 = |Q|^2 + |R|^2 - 2r^2$, which is equivalent to $Q \cdot R = r^2$. That says R lies on the polar of Q, which is Brocard's theorem. L5-GE2 proves this identity again with the Miquel point.

1.5 5. Projective maps

Write the point (x, y) as $[x : y : 1]$, where $[x : y : z] = [\lambda x : \lambda y : \lambda z]$ for $\lambda \neq 0$; the points $[x : y : 0]$ form the line at infinity. A *projective transformation* is a map $[v] \mapsto [Mv]$ with M an invertible 3×3 matrix. It maps lines to lines and preserves cross-ratios (Chapter 9 of Evan Chen's *Euclidean Geometry in Mathematical Olympiads* develops this).

Theorem 5.1.
(1) For any line ℓ there is a projective transformation sending ℓ to the line at infinity. (2) For any circle ω and point P inside it, there is a projective transformation sending ω to a circle

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and P to its centre.

Proof. (1) If ℓ is $ax + by + cz = 0$, take M with third row (a, b, c) and two further rows making it invertible; then ℓ goes to $z = 0$. (2) Similarities are projective, so we may take ω to be $x^2 + y^2 = z^2$ and $P = [p : 0 : 1]$ with $p \in [0, 1)$. The map $[x : y : z] \mapsto [x - pz : \sqrt{1 - p^2}\, y : z - px]$ multiplies $x^2 + y^2 - z^2$ by $1 - p^2$, so it maps ω to itself, and it sends P to $[0 : 0 : 1 - p^2]$, the centre. ■

As an illustration, here is a second proof of the quadrilateral harmonic range $(A, D; X, R) = -1$. Send line QR to infinity. Then $AB \parallel CD$ and $AD \parallel BC$, so $ABCD$ becomes a parallelogram and P its centre. Line QP is now the line through the centre parallel to AB, so X is the midpoint of AD, while R is at infinity; hence $(A, D; X, R) = -1$, and cross-ratio is preserved.

Common traps.

- Assuming that the centre of a circle is mapped to the centre of its image.
- Forgetting the special cases of inversion: lines and circles through the centre, and the centre itself.
- Dropping signs in a cross-ratio, or forgetting that a harmonic conjugate can be the point at infinity.
- Using a projective map to deduce a statement about lengths, angles or perpendicularity.

1.6 How to spot this in a contest

1. Several circles through one point, or two circles touching at a point: invert there.
2. Tangents from a point together with a secant, or a cyclic quadrilateral with the intersections of opposite sides: think of polars and Brocard.
3. A midpoint together with a parallel line: a harmonic range with a point at infinity.
4. Incircle touch points: the cevians to them concur, so Theorem 3.1(2) applies at once.

Chapter 4 of my book *Beyond the Incenter* treats cross-ratios at greater length, including harmonic bundles, perspectivity and Pascal's theorem.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Circles of radii 1 and 2 touch externally at T, and ℓ is the line through their centres. Find the radius of a circle which touches both circles and ℓ.

Problem 2 ⋆⋆ Let A, B, C, D be four points, no three collinear and not all on one circle. The angle between two intersecting circles means the acute or right angle between their tangents at a common point. Prove that the angle between circles ABC and ABD equals the angle between circles CDA and CDB.

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Problem 3 ⋆⋆⋆ Let $ABCD$ be a convex quadrilateral with no two sides parallel. Let $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$. Prove that $\angle PRQ = 90^\circ$ if and only if RP bisects angle ARB.

Problem 4 ⋆⋆⋆ Circles ω_1 and ω_2 meet at two points A and B. A circle Γ touches ω_1 at P and ω_2 at Q, where A, B, P, Q are four different points. Prove that $PA \cdot QB = PB \cdot QA$.

Problem 5 ⋆⋆⋆ Four different circles $\omega_1, \omega_2, \omega_3, \omega_4$ pass through a point A. Circles ω_1 and ω_3 touch at A, circles ω_2 and ω_4 touch at A, and the two common tangents at A are different lines. Circles ω_1 and ω_2 meet again at P, circles ω_2 and ω_3 at Q, circles ω_3 and ω_4 at R, and circles ω_4 and ω_1 at S. Prove that

$$AP^2 \cdot QR \cdot RS = AR^2 \cdot PQ \cdot PS.$$

Problem 6 ⋆⋆⋆⋆ Triangle ABC has circumcentre O, circumradius R, incentre I and inradius r. The incircle touches BC, CA, AB at D, E, F, and H is the orthocentre of triangle DEF. Prove that

$$\overrightarrow{IH} = -\frac{r}{R}\overrightarrow{IO},$$

so that H lies on line OI (when $O \neq I$), and deduce Euler's formula $OI^2 = R^2 - 2Rr$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **BMO2 2002/03 Problem 2** ⋆⋆
   Find two similar triangles sharing the angle at A; then compare AP squared with AD times AB.
2. **BMO2 2017/18 Problem 1** ⋆⋆
   Invert at B with power BA times BP and ask where M goes.
3. **BMO2 2013/14 Problem 4** ⋆⋆⋆
   Find the direction of the tangent at P to circle BCP.
4. **IMO 2003 Problem 4** ⋆⋆⋆
   Turn both conditions into statements about DA/DC and BA/BC.
5. **IMO 2010 Problem 4** ⋆⋆⋆
   The circle centred at S through C is orthogonal to the circumcircle; it is an Apollonius circle.
6. **IMO 1996 Problem 2** ⋆⋆⋆
   Invert at A, or use the inversion distance formula, to turn the angle condition into a length condition.
7. **USAMO 2015 Problem 2** ⋆⋆⋆
   Show that AX times AS is constant; inversion at A then swaps X and S.
8. **IMO 2015 Problem 3** ⋆⋆⋆⋆
   Invert at H with the negative power that swaps the circumcircle and the nine-point circle.
9. **USAMO 2017 Problem 3** ⋆⋆⋆⋆
   Find an inversion centred at S that swaps B and C and also swaps K and M.

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4 Hints

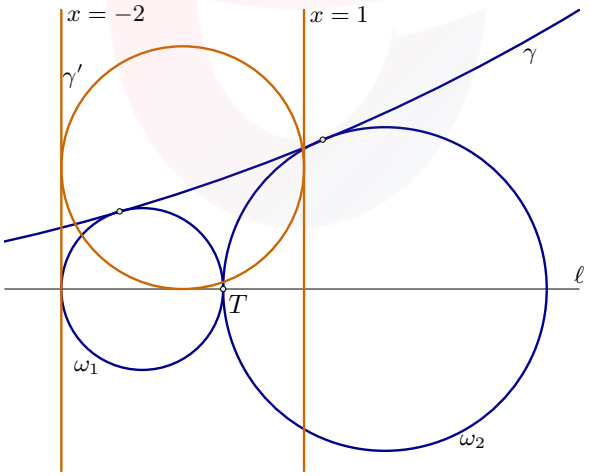
1. Invert at T. What do the two circles and ℓ become?
2. Three of the four circles pass through A.
3. Find a harmonic pencil at R containing the lines RP and RQ.
4. Invert at A. What can you say about the two tangent lengths from B' to Γ'?
5. Invert at A. What do two circles touching at the centre of inversion become, and what quadrilateral do the four images form?
6. Invert in the incircle. The image of A is a point you know in triangle DEF, and the image of the circumcircle is a circle you know.

5 Solutions

Problem 1

Circles of radii 1 *and* 2 *touch externally at* T*, and* ℓ *is the line through their centres. Find the radius of a circle which touches both circles and* ℓ*.*

Use coordinates with T at the origin and ℓ the x-axis, so the centres are $(-1,0)$ and $(2,0)$. Invert at T with $r^2 = 4$. The first circle passes through T and $(-2,0)$, which is fixed, so it becomes the line $x = -2$ (perpendicular to ℓ, by symmetry). The second passes through T and $(4,0)$, whose image is $(1,0)$, so it becomes the line $x = 1$. The line ℓ passes through T and is fixed.



A circle γ touching both circles and ℓ cannot pass through T: it would have to touch both circles at T, so its tangent at T would be the common tangent, which is perpendicular to ℓ, and then γ would cross ℓ at T instead of touching it. So γ' is a circle touching $x = -2$, $x = 1$ and ℓ (Theorem 1.3). It therefore has radius $\frac{3}{2}$ and centre $(-\frac{1}{2}, \pm\frac{3}{2})$; by symmetry take the plus sign.

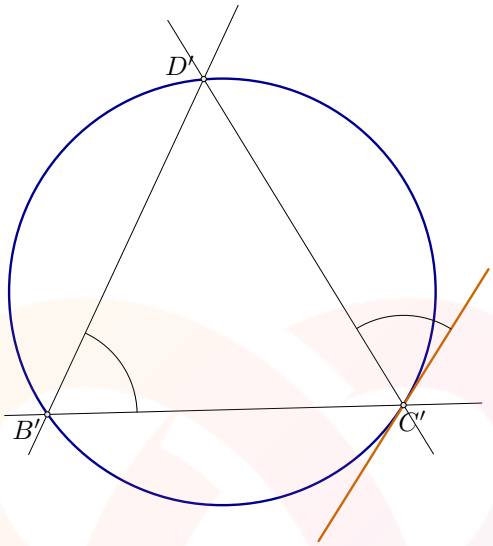
The power of T with respect to γ' is $k = \frac{1}{4} + \frac{9}{4} - \frac{9}{4} = \frac{1}{4}$. By the proof of Theorem 1.2(3), γ is the image of γ' under the homothety at T with ratio $r^2/k = 16$, so its radius is $16 \cdot \frac{3}{2} = 24$ and its centre is $(-8, 24)$. As a check, the distances from $(-8,24)$ to $(-1,0)$ and $(2,0)$ are $25 = 24 + 1$ and $26 = 24 + 2$. There are two such circles, one on each side of ℓ, both of radius 24.

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Answer: 24

Problem 2

Let A, B, C, D be four points, no three collinear and not all on one circle. The angle between two intersecting circles means the acute or right angle between their tangents at a common point. Prove that the angle between circles ABC and ABD equals the angle between circles CDA and CDB.



Invert at A with any radius. Circles ABC, ABD and CDA pass through A, so they become the lines $B'C'$, $B'D'$ and $C'D'$. Circle CDB does not pass through A, since the points are not concyclic, so it becomes the circle $C'D'B'$. The points B', C', D' are not collinear, since otherwise circle BCD would pass through A.

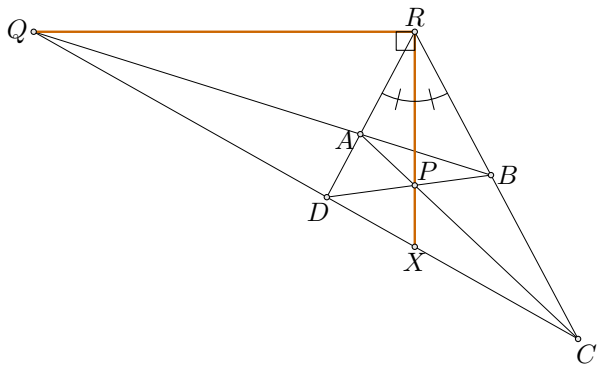
By Theorem 1.3, the angle between circles ABC and ABD at their common point B equals the angle between lines $B'C'$ and $B'D'$, which is the acute or right angle determined by $\angle C'B'D'$. The angle between circles CDA and CDB at their common point C equals the angle at C' between the line $C'D'$ and the circle $C'D'B'$, that is, between the chord $C'D'$ and the tangent at C'. By the alternate segment theorem this angle is $\angle C'B'D'$ or its supplement, so it determines the same acute or right angle. Hence the two angles are equal.

Problem 3

Let $ABCD$ be a convex quadrilateral with no two sides parallel. Let $P = AC \cap BD$, $Q = AB \cap CD$ and $R = AD \cap BC$. Prove that $\angle PRQ = 90^\circ$ if and only if RP bisects angle ARB.

The relabelling $A \leftrightarrow D$, $B \leftrightarrow C$ leaves P, Q, R unchanged, and angle ARB is the same as angle DRC, since by convexity A, D lie on one ray from R and B, C on another. So we may assume A lies between R and D, and then, by convexity, B lies between R and C, and the quadrilateral lies inside triangle RDC.

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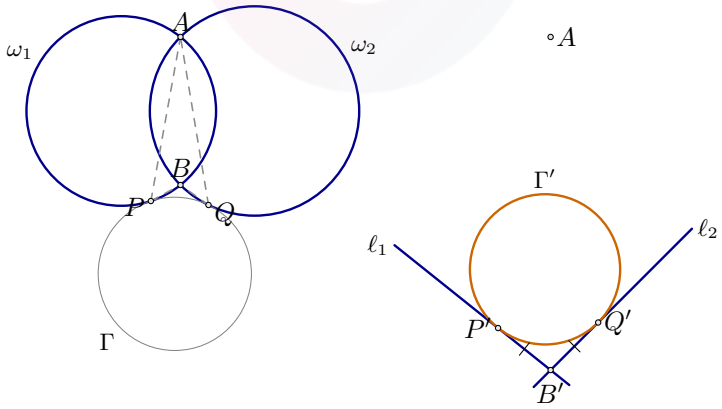
In triangle RDC the points A and B lie on sides RD and RC, and the cevians DB and CA meet at P. Let RP meet DC at X; since P is inside the triangle, X lies on segment DC. The line AB meets DC at Q. By Theorem 3.1(2), $(D, C; X, Q) = -1$. As Q is a finite point, X is not the midpoint of DC.

Suppose $\angle PRQ = 90°$. Then $\angle XRQ = 90°$ and $(D, C; X, Q) = -1$, so by Theorem 3.1(1) the lines RX and RQ bisect the angles between RD and RC. Since X lies on segment DC, RX is the internal bisector of angle DRC, that is, RP bisects angle ARB.

Conversely, suppose RP bisects angle $ARB = DRC$, so RX is the internal bisector. Then $RD/RC = XD/XC \neq 1$, so the external bisector at R is not parallel to DC and meets it at a point Q_1 with $(D, C; X, Q_1) = -1$ by Theorem 3.1(1). The harmonic conjugate of X with respect to D, C is unique, so $Q_1 = Q$. Hence RQ is the external bisector, which is perpendicular to the internal bisector RP, and $\angle PRQ = 90°$.

Problem 4

Circles ω_1 and ω_2 meet at two points A and B. A circle Γ touches ω_1 at P and ω_2 at Q, where A, B, P, Q are four different points. Prove that $PA \cdot QB = PB \cdot QA$.



Invert at A with radius 1. Since ω_1 and ω_2 pass through A and B, their images are two different lines ℓ_1, ℓ_2 through B'. The circle Γ does not pass through A: the only common point of Γ and ω_1 is P, and $P \neq A$. So Γ' is a circle, and by Theorem 1.3 it touches ℓ_1 at P' and ℓ_2 at Q'. Similarly B is not on Γ (as $B \neq P$), so B' is not on Γ'.

Hence $B'P'$ and $B'Q'$ are the two tangent segments from B' to Γ', and $B'P' = B'Q'$. By Theorem 1.1,

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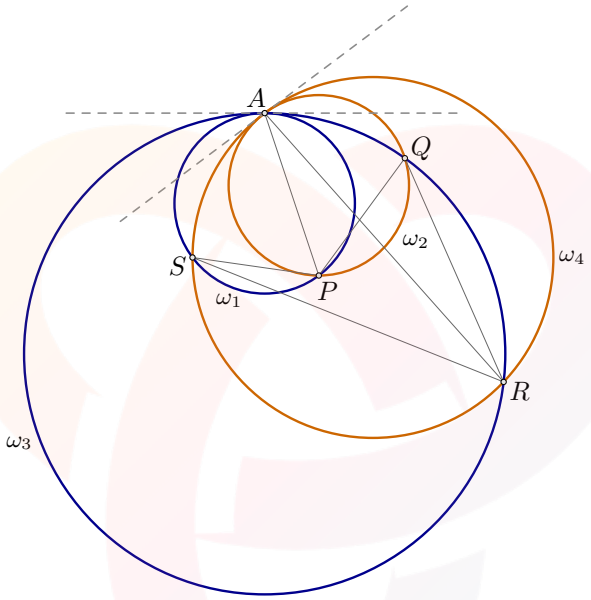
$$B'P' = \frac{BP}{AB \cdot AP}, \qquad B'Q' = \frac{BQ}{AB \cdot AQ}.$$

Equating and multiplying by $AB \cdot AP \cdot AQ$ gives $BP \cdot AQ = BQ \cdot AP$, as required. In other words, P and Q lie on the same Apollonius circle of A and B.

Problem 5

Four different circles $\omega_1, \omega_2, \omega_3, \omega_4$ pass through a point A. Circles ω_1 and ω_3 touch at A, circles ω_2 and ω_4 touch at A, and the two common tangents at A are different lines. Circles ω_1 and ω_2 meet again at P, circles ω_2 and ω_3 at Q, circles ω_3 and ω_4 at R, and circles ω_4 and ω_1 at S. Prove that

$$AP^2 \cdot QR \cdot RS = AR^2 \cdot PQ \cdot PS.$$



First the points exist. The circles ω_1 and ω_2 have different tangents at A, so they do not touch there and meet at a second point $P \neq A$. Similarly Q, R and S exist and differ from A.

Invert at A with radius 1. By Theorem 1.2 each ω_i is mapped to a line ℓ_i not through A, parallel to the tangent of ω_i at A. Two circles that touch at A have no common point other than A, which goes to ∞, so ℓ_1 and ℓ_3 have no common point: they are different parallel lines. Similarly $\ell_2 \parallel \ell_4$. As the tangents of ω_1 and ω_2 at A are different, ℓ_1 and ℓ_2 are not parallel. The second intersection P of ω_1 and ω_2 goes to the only common point of ℓ_1 and ℓ_2, so $P' = \ell_1 \cap \ell_2$, and in the same way $Q' = \ell_2 \cap \ell_3$, $R' = \ell_3 \cap \ell_4$, $S' = \ell_4 \cap \ell_1$.

Hence $P'Q'R'S'$ is a parallelogram whose sides $P'Q'$, $Q'R'$, $R'S'$, $S'P'$ lie on ℓ_2, ℓ_3, ℓ_4, ℓ_1; its four vertices are different, since any two of them lie on ℓ_1 and ℓ_3 respectively, or on ℓ_2 and ℓ_4, and these pairs of lines are disjoint. Opposite sides of a parallelogram are equal, so $P'Q' = R'S'$ and $Q'R' = S'P'$.

Now Theorem 1.1 (which also holds when A and the two points are collinear) gives $P'Q' = \frac{PQ}{AP \cdot AQ}$, and similarly for the other three sides. The two equalities become

$$\frac{PQ}{AP \cdot AQ} = \frac{RS}{AR \cdot AS'}, \qquad \frac{QR}{AQ \cdot AR} = \frac{SP}{AS \cdot AP}.$$

Dividing the first by the second, AQ and AS cancel and we get

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$$\frac{PQ \cdot AR}{QR \cdot AP} = \frac{RS \cdot AP}{SP \cdot AR},$$

that is, $AP^2 \cdot QR \cdot RS = AR^2 \cdot PQ \cdot PS$.

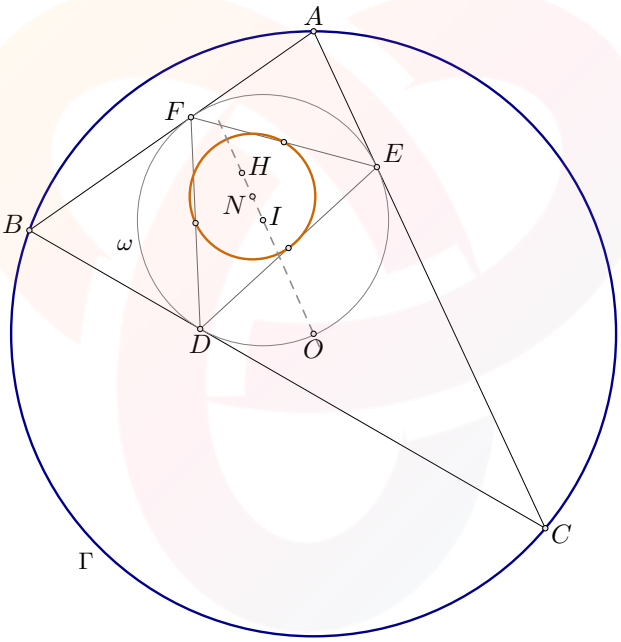
Each displayed equality is a result on its own, for example $PQ \cdot AR \cdot AS = RS \cdot AP \cdot AQ$, and the problem needs both. In problems of this kind, write down every simple fact about the inverted figure (here: opposite sides parallel and equal) and translate each one back with Theorem 1.1.

Problem 6

Triangle ABC has circumcentre O, circumradius R, incentre I and inradius r. The incircle touches BC, CA, AB at D, E, F, and H is the orthocentre of triangle DEF. Prove that

$$\overrightarrow{IH} = -\frac{r}{R}\,\overrightarrow{IO},$$

so that H lies on line OI (when $O \neq I$), and deduce Euler's formula $OI^2 = R^2 - 2Rr$.



Let ω be the incircle and invert in ω. The point A lies outside ω with tangents AE and AF, so its inverse is the point where AI meets the chord of contact EF (Section 4). As $AE = AF$ and $IE = IF$, line AI is the perpendicular bisector of EF, so this point is the midpoint of EF. Similarly B and C go to the midpoints of FD and DE. The circumcircle Γ does not pass through I (which is inside it), so by Theorem 1.2 its image Γ' is the circle through the three midpoints of the sides of DEF, that is, the nine-point circle of DEF.

We use the standard facts about the nine-point circle of a triangle: its radius is half the circumradius, and its centre is the midpoint of the segment joining the circumcentre to the orthocentre. The circumcircle of DEF is ω, with centre I and radius r, so Γ' has radius $r/2$ and centre N, the midpoint of IH.

Now compute Γ' from the proof of Theorem 1.2(3). The power of I with respect to Γ is $k = OI^2 - R^2$, which is negative as I is inside Γ. So Γ' is the image of Γ under the homothety with centre I and ratio r^2/k. Comparing radii, $R \cdot r^2/|k| = r/2$, so $|k| = 2Rr$, that is, $R^2 - OI^2 = 2Rr$, which is Euler's formula. Comparing centres,

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$$\overrightarrow{IN} = \frac{r^2}{k}\overrightarrow{IO} = -\frac{r^2}{2Rr}\overrightarrow{IO} = -\frac{r}{2R}\overrightarrow{IO}.$$

Since N is the midpoint of IH, $\overrightarrow{IH} = 2\overrightarrow{IN} = -\frac{r}{R}\overrightarrow{IO}$. If $O \neq I$ this puts H on line OI, on the other side of I from O; if $O = I$ (the equilateral case) it says $H = I$, which is also true.